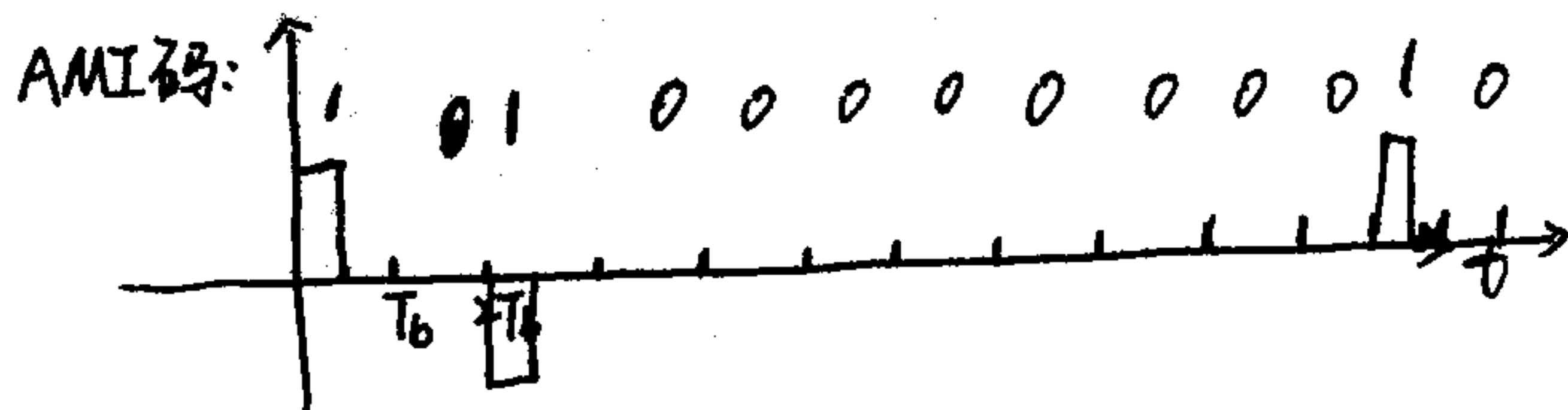
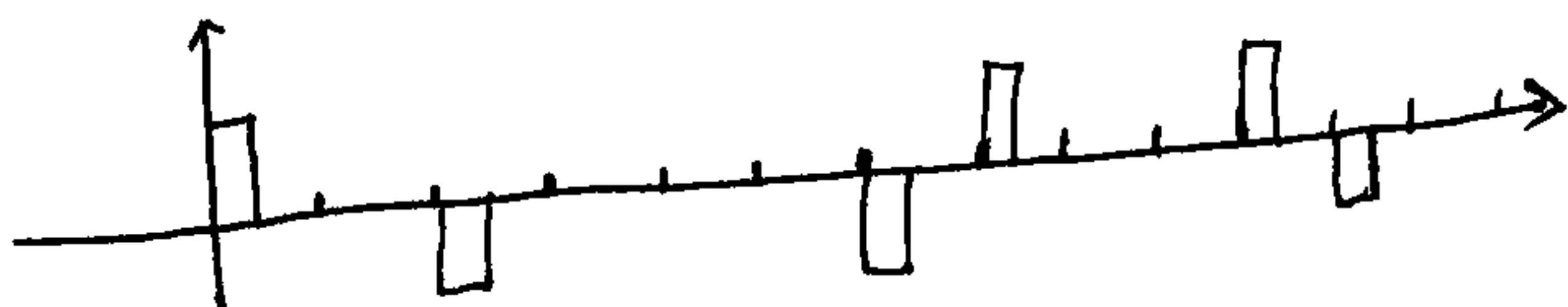


5-2. 原序列: 101000000010



HDB3: +B 0 -B 000 -V +B' 0 0 +V -B 0
+1 0 -1 000 -1 +1 0 0 +1 -1 0



5-5. (1) $\{a_n\}$ 独立不相关, 其功率谱密度满足:

$$P_s(f) = \frac{\sigma_a^2}{T_s} |G_T(f)|^2 + \frac{m_a^2}{T_s^2} \sum_{m=-\infty}^{\infty} |G_T(\frac{m}{T_s})|^2 \delta(f - \frac{m}{T_s})$$

解: $m_a=0$, $\sigma_a^2 = \frac{1}{8} [(1+3^2+5^2+7^2) \times 2] = 21$

又: $|G_T(f)| = T_s \text{Sa}(\pi f T_s)$

$\therefore P_s(f) = 21 T_s \text{Sa}^2(\pi f T_s)$

(2) $m_a=0$, $\sigma_a^2 = \frac{1}{2}(1+1) = 1$

$\therefore P_s(f) = \frac{\sigma_a^2}{T_s} |G_T(f)|^2 = \frac{1}{T_s} \cdot T_s^2 \cdot \text{Sa}^2(\pi f T_s) = T_s \text{Sa}^2(\pi f T_s)$

(3). $T = \frac{1}{T_s} = R_s = 10^6$

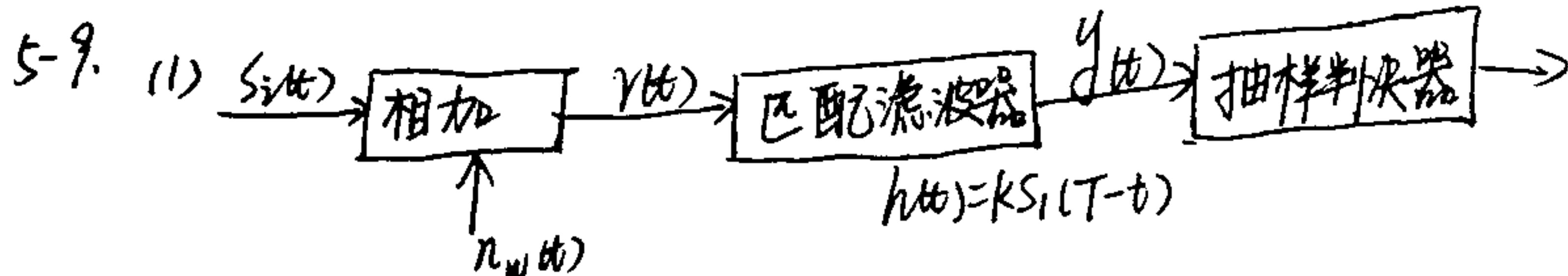
$\therefore T_{2PAM} = 10^6 \text{ Hz}$

$T_{8PAM} = 10^6 \text{ Hz}$

(4). $R_b = 6 \times 10^6 \text{ bit/s}$

对于 2PAM: $R_s = R_b / \log_2 2 = R_b = 6 \times 10^6 \text{ Band}$, $\therefore T_{2PAM} = 6 \times 10^6 \text{ Hz}$

对于 8PAM: $R_s = R_b / \log_2 8 = 2 \times 10^6 \text{ Band}$, $\therefore T_{8PAM} = 2 \times 10^6 \text{ Hz}$

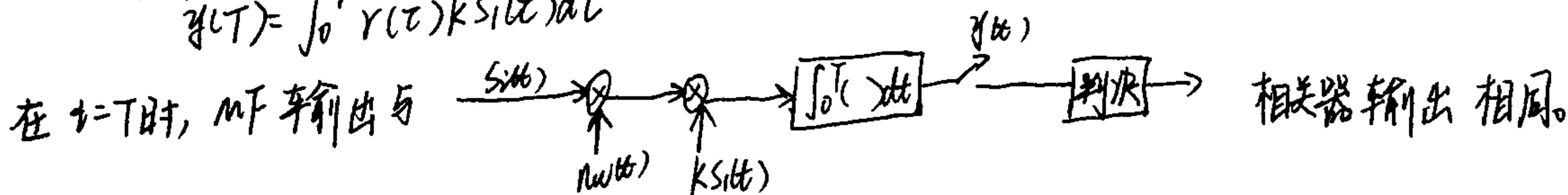


匹配滤波器输出表示:

$$y(t) = \int_{-\infty}^{\infty} r(\tau) h(t-\tau) d\tau = \int_0^t r(\tau) h(t-\tau) d\tau = \int_0^t r(\tau) k s_i(T-\tau+\tau) d\tau$$

当 $t=T$ 时:

$$y(T) = \int_0^T r(\tau) k s_i(\tau) d\tau$$



①

(2). 设 $k=1$, 则抽样时刻 $y(t)$ 的信号幅度为: $A^2 T$

信号的瞬时功率: $P = (A^2 T)^2 = A^4 T^2$

噪声的平均功率: $S = \frac{N_0}{2} A^2 T$

输出信噪比: $\gamma = \frac{A^4 T^2}{\frac{N_0}{2} A^2 T} = \frac{2A^2 T}{N_0}$

(3). 当 $k = \frac{1}{\sqrt{E}}$, $E = A^2 T$, 则抽样时刻 $y(t)$ 的信号幅度为: $\sqrt{E} = \sqrt{A^2 T}$

信号瞬时功率: $P = E = A^2 T$, 噪声平均功率: $S = \frac{N_0}{2}$,

输出信噪比: $\gamma = \frac{A^2 T}{\frac{N_0}{2}} = \frac{2}{N_0} A^2 T = \frac{2E}{N_0}$

(4). 当 $P(s_1) = P(s_2) = \frac{1}{2}$ 时, $P_b = Q(\sqrt{\frac{2A^2 T}{N_0}})$

5-10. 根据 $Q(\sqrt{\frac{2E_b}{N_0}})$ 图像, 若 $P_b \leq 10^{-6}$

$$\text{则 } \left(\frac{E_b}{N_0}\right)_{\text{dB}} \geq 10.5 \Rightarrow \frac{E_b}{N_0} \geq 10^{1.05} \approx 11.23$$

$$\therefore E_b \geq 11.23 \times 10^{-10} = 1.13 \times 10^{-9}$$

$$\therefore \text{接收信号的平均功率: } P = E_b R_b = 1.13 \times 10^{-9} \times 10^6 = 1.13 \times 10^{-3} \text{ W/Hz}$$

5-15. 对于(a). $B_N = 1500$, $R_s = 2B_N = 3000 \text{ Baud}$,
则对于该 $X(f)$, 满足无码间干扰条件为 $R = \frac{R_s}{K} = \frac{3000}{K}$ (K 为整数)

$\therefore 1000, 3000 \text{ Baud}$ 满足无码间干扰

$\therefore$ 最大频带利用率: $\frac{R_s}{B} = \frac{3000}{2000} = 1.5 \text{ Baud/Hz}$

对于(b). $B_N = 2000$, $R_s = 2B_N = 4000 \text{ Baud}$

则对于该 $X(f)$, 满足无码间干扰条件为 $R = \frac{4000}{K}$ (K 为整数)

$\therefore 1000, 2000, 4000 \text{ Baud}$ 能无码间干扰

最大频带利用率: $\frac{R_s}{B} = \frac{4000}{3000} = \frac{4}{3} \text{ Baud/Hz}$

5-18.

(1) $\because B = 640 \text{ KHz}$,

$$\alpha = \frac{B - B_N}{B_N} = 1$$

$$\therefore B_N = 320 \text{ KHz}$$

系统最大信息传输速率 $R_s = 2B_N = 640 \text{ K Baud/s}$.

(2). 每路信号最大信息传输速率: $\frac{640 \text{ K}}{10} = 64 \text{ K Baud/s}$

又: 每路信号经过 A 律 13 折线码编, $\therefore$ 采样频率: $f_s = \frac{64 \text{ K}}{8} = 8 \text{ K}$

又根据奈奎斯特编码定理 $f_s = 2f_H \Rightarrow f_H = 4 \text{ K (Hz)}$

$$5-21. (1) \because \alpha = \frac{B - B_N}{B_N} \text{ 则 } 0.15 = \frac{3000 - B_N}{B_N} \Rightarrow B_N = 2400 \text{ Hz}$$

$$\therefore (R_s)_{\text{max}} = 2B_N = 4800 \text{ Baud/s}$$

$$R_b = \log_2 M \cdot R_s = 4800 \times 3 = 14400 \text{ bit/s}$$

(2)

$$(2). \because d = \frac{B - B_N}{B_N} \quad \therefore 0.5 = \frac{3000 - B_N}{B_N} \Rightarrow B_N = 2000$$

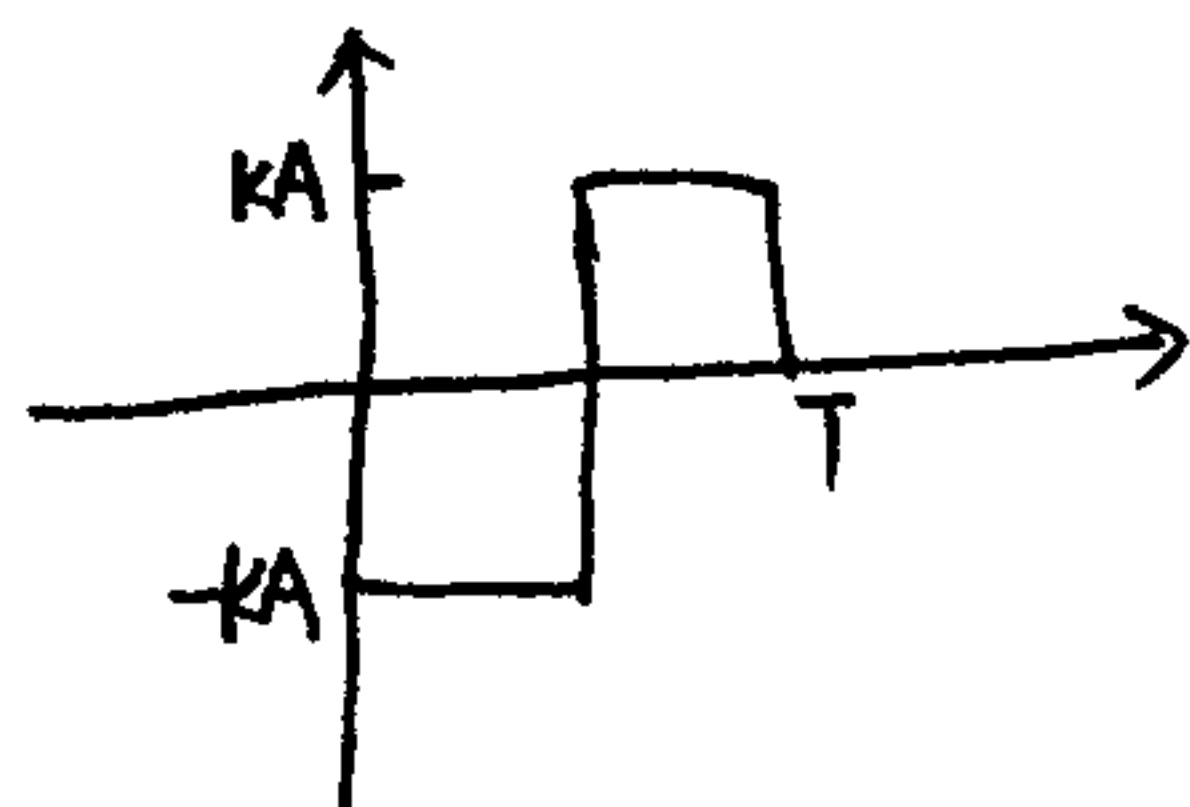
$$\therefore (R_s)_{\max} = 2B_N = 4000 \text{ Band}$$

$$R_b = \log_2 16 \cdot R_s = 4 \times 4000 = 16000 \text{ bit/s}$$

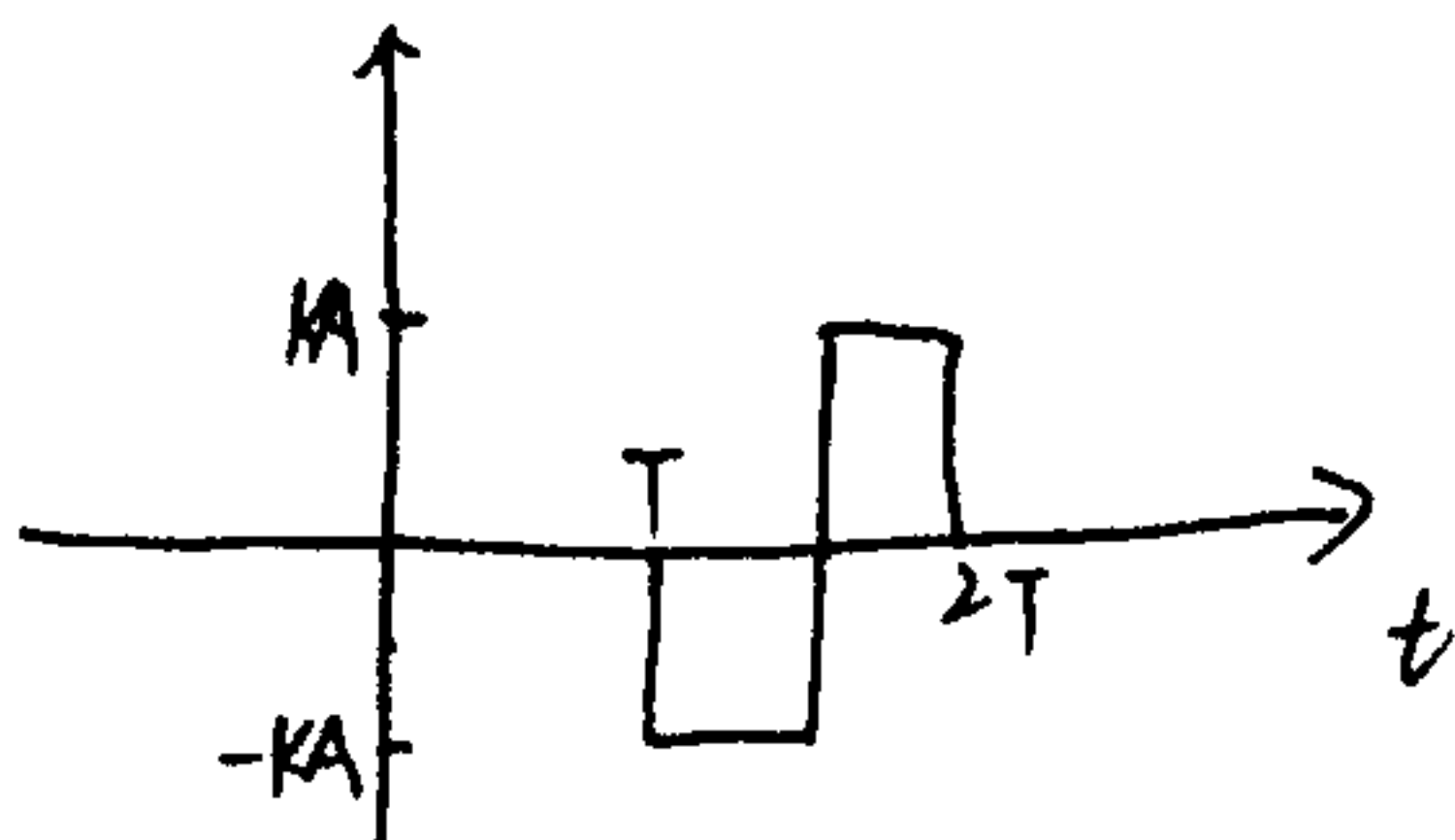
2-22.

(1) 当 $t = T$ 时

$$h(t) = KS(T - t)$$



当 $t = 2T$ 时 $h(t) = KS(2T - t)$



(2). $t \geq T$

(3). 抽样时刻信号瞬时值: A^2T

$$\text{瞬时功率: } (A^2T)^2 = A^4T^2$$

$$\text{噪声功率: } \frac{N_0}{2} A^2T$$

$$\text{最大输出信噪比: } \gamma = \frac{A^4T^2}{\frac{N_0}{2} A^2T} = \frac{2A^2T}{N_0}$$

$$\text{噪声均值为0, 方差为 } \frac{N_0}{2} A^2T = \sigma^2$$

$$\text{则噪声的一维概率密度函数: } f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{x^2}{2\sigma^2}\right]$$

(4). 当 $k = \frac{1}{\sqrt{E}}$ 时, 信号瞬时值: $\sqrt{A^2T}$

$$\text{信号瞬时功率: } P = A^2T$$

$$\text{最大输出信噪比: } \gamma = \frac{2A^2T}{N_0}$$

$$\text{噪声均值为0, 方差为 } \frac{N_0}{2} = \sigma^2$$

$$\therefore f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{x^2}{N_0}\right]$$